

若干点电荷 q_i 和若干导体 Q_i 构成的系统

$$W = \frac{1}{2} \iiint \rho \varphi dV = \frac{1}{2} \sum Q_i \varphi_i + \frac{1}{2} \sum q_i \varphi_i$$

其中 φ_i 、 φ_i' 分别为导体 Q_i 和点电荷 q_i 处的总电势。

$$\varphi_i = \varphi_{i0} + \varphi_i' \quad \begin{cases} \varphi_{i0} \text{ 为点电荷 } q_i \text{ 在自身处的电势} \\ \varphi_i' \text{ 为其他电荷在 } q_i \text{ 处的电势。} \end{cases}$$

φ_{i0} 发散，且对静电能的贡献不随点电荷位置的变化而变化。

重新定义点电荷-导体系统的静电能

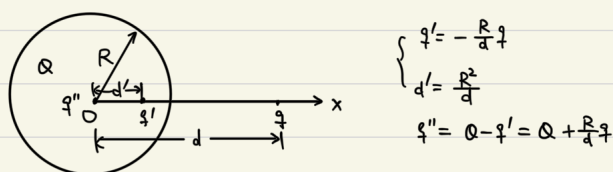
$$W = \frac{1}{2} \sum Q_i \varphi_i + \frac{1}{2} \sum q_i \varphi_i'$$

方法一：直接求静电力 $\vec{F} = q\vec{E}$

方法二：由静电能求力 $W = \frac{1}{2} Q\varphi + \frac{1}{2} q\varphi'$ $\vec{F} = -\nabla W$

方法三：由电势能求力 $U = q\varphi'$ $\vec{F} = -\nabla U$
(对应的能量无实际意义，只是数学工具) (此 ∇ 非彼 ∇ ，求导的条件不同)

例1. 半径为 R 、带电量为 Q 的导体球固定，距离球心 d ($d > R$) 处有点电荷 q ，求 q 受到的静电力。



方法一：直接求静电力。

$$\vec{F} = q \frac{1}{4\pi\epsilon_0} \left(\frac{-\frac{R}{d}q}{(d - \frac{R^2}{d})^2} + \frac{Q + \frac{R}{d}q}{d^2} \right) \hat{x}$$

方法二：由静电能求力

$$W = \frac{1}{2} Q\varphi + \frac{1}{2} q\varphi'$$

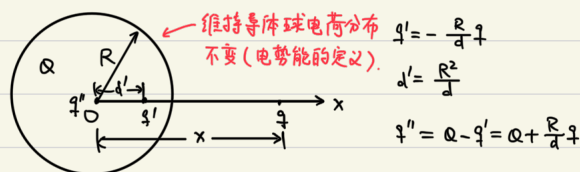
$$\varphi = \frac{q''}{4\pi\epsilon_0 R} = \frac{Q + \frac{R}{d}q}{4\pi\epsilon_0 R}$$

$$\begin{aligned} \varphi' &= \frac{1}{4\pi\epsilon_0} \left(\frac{q''}{x} + \frac{q'}{x - x'} \right) & \begin{cases} q' = -\frac{R}{d}q \\ x' = \frac{R^2}{d} \end{cases} \\ &= \frac{1}{4\pi\epsilon_0} \left(\frac{Q + \frac{R}{d}q}{x} + \frac{-\frac{R}{d}q}{x - \frac{R^2}{d}} \right) & q'' = Q - q' = Q + \frac{R}{d}q \\ &= \frac{1}{4\pi\epsilon_0} \left(\frac{Q}{x} + \frac{Rq}{x^2} - \frac{Rq}{x^2 - R^2} \right) \end{aligned}$$

$$\begin{aligned} W &= \frac{1}{8\pi\epsilon_0} \left(\frac{Q^2}{R} + \frac{qQ}{x} + \frac{qQ}{x} + \frac{Rq^2}{x^2} - \frac{Rq^2}{x^2 - R^2} \right) \\ &= \frac{1}{8\pi\epsilon_0} \left(\frac{Q^2}{R} + \frac{2qQ}{x} + \frac{Rq^2}{x^2} - \frac{Rq^2}{x^2 - R^2} \right) \end{aligned}$$

$$\begin{aligned} \vec{F} &= -\nabla W = \frac{1}{8\pi\epsilon_0} \left(\frac{2qQ}{x^2} + \frac{2Rq^2}{x^3} - \frac{Rq^2 \cdot 2x}{(x^2 - R^2)^2} \right) \Big|_{x=d} \hat{x} \\ &= \frac{q}{4\pi\epsilon_0} \left(\frac{Q}{d^2} + \frac{Rq}{d^3} - \frac{Rd}{(d^2 - R^2)^2} \right) \hat{x} \end{aligned}$$

方法三：由电势能求力。



$$\varphi' = \frac{1}{4\pi\epsilon_0} \left(\frac{-\frac{R}{d}q}{x - \frac{R^2}{d}} + \frac{Q + \frac{R}{d}q}{x} \right)$$

$$U = q\varphi' = \frac{q}{4\pi\epsilon_0} \left(\frac{-\frac{R}{d}q}{x - \frac{R^2}{d}} + \frac{Q + \frac{R}{d}q}{x} \right)$$

$$\begin{aligned} \vec{F} &= -\nabla U = \frac{q}{4\pi\epsilon_0} \left(\frac{-\frac{R}{d}q}{(x - \frac{R^2}{d})^2} + \frac{Q + \frac{R}{d}q}{x^2} \right) \Big|_{x=d} \hat{x} \\ &= \frac{q}{4\pi\epsilon_0} \left(\frac{-Rd}{(d^2 - R^2)^2} + \frac{Q}{d^2} + \frac{Rq}{d^3} \right) \hat{x} \end{aligned}$$

例2. 内径为 R 、外径为 $2R$ 的不带导体球壳，在过球心的一条直线上放置两个点电荷。一个点电荷电量为 $q_1 = q$ ，到球心的距离为 $d_1 = \frac{R}{2}$ 。另一个点电荷电量为 $q_2 = q$ ，到球心的距离为 $d_2 = (\sqrt{5} + 1)R$ 。

(1) 求点电荷 q_1, q_2 的受力 F_1, F_2 。

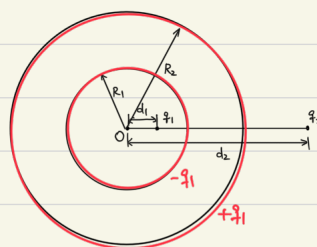
(2) 求球壳的电势 V 和球心处的电势 V_0 。

(3) 分别求导体球壳内表面感应电荷的自能和导体球壳内表面感应电荷与点电荷 q_1 之间的相互作用能。

(4) 分别求导体球壳外表面感应电荷的自能和导体球壳外表面感应电荷与点电荷 q_2 之间的相互作用能。

(5) 维持导体电荷量不变。将点电荷 q_2 无限缓慢地移至无穷远处，作用于点电荷 q_2 上用以抵抗静电力的外力需做多大的功？

(5) 维持导体电势 V 不变。将点电荷 q_2 无限缓慢地移至无穷远处，作用于点电荷 q_2 上用以抵抗静电力的外力需做多大的功？



$$q_1' = -\frac{R}{d_1} q_1 = -2q$$

$$d_1' = \frac{R^2}{d_1} = 2R$$

$$q_2' = -\frac{2R}{d_2} q_2 = -\frac{\sqrt{5}-1}{2} q$$

$$d_2' = \frac{(2R)^2}{d_2} = (\sqrt{5}-1)R$$

$$q_2'' = q_1 - q_1' = \frac{\sqrt{5}+1}{2} q$$

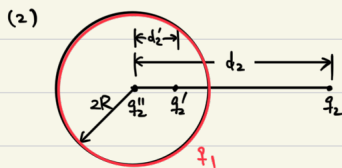
(1) 对 q_1 : 由于静电屏蔽, 外表面电荷 q_2 对球壳内电场无影响.

q_1 受力仅由内表面感应电荷引起.

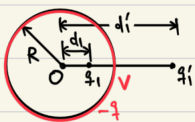
$$\vec{F}_1 = q_1 \frac{1}{4\pi\epsilon_0} \frac{q'_1}{(d'_1 - d)^2} (-\hat{x}) = \frac{q \cdot (-2q)}{4\pi\epsilon_0 (2R - \frac{R}{2})^2} (-\hat{x}) = \frac{2q^2}{9\pi\epsilon_0 R^2} \hat{x}$$

方向: 沿半径向外

$$\begin{aligned} \text{对 } q_2: \vec{F}_2 &= q \cdot \frac{1}{4\pi\epsilon_0} \left[\frac{q'_2}{(d_2 - d'_2)^2} + \frac{q''_2}{d_2^2} \right] \hat{x} \\ &= q \cdot \frac{1}{4\pi\epsilon_0} \left[\frac{-\frac{\sqrt{5}-1}{2}q}{(2R)^2} + \frac{\frac{\sqrt{5}+1}{2}q}{((\sqrt{5}+1)R)^2} \right] \hat{x} \\ &= 0 \end{aligned}$$



$$V = \frac{q''_2}{4\pi\epsilon_0 (2R)} = \frac{\frac{\sqrt{5}+1}{2}q}{4\pi\epsilon_0 \cdot (2R)} = \frac{\sqrt{5}+1}{4} \frac{q}{4\pi\epsilon_0 R}$$



$$V_0 = V + \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{d_1} + \frac{q'_1}{d'_1} \right) = V + \frac{q}{4\pi\epsilon_0 R} = \frac{\sqrt{5}+5}{4} \frac{q}{4\pi\epsilon_0 R}$$

(3) 球壳内表面的感应电荷的自能, 只由其电荷分布决定.

$$W_{1a} = \frac{1}{2} \oint \tilde{\varphi}'_1 dq'_1$$

其中 φ'_1 为内表面感应电荷在自身处电势.

$$0 = \tilde{\varphi}'_1 + \varphi_1$$

其中 φ_1 为 q_1 在内表面的电势

$$W_{1a} = -\frac{1}{2} \oint \varphi_1 dq'_1 \quad \begin{matrix} \text{格林} \\ \text{互易定理} \end{matrix} = -\frac{1}{2} q_1 \varphi'_1$$

其中 φ'_1 为内表面感应电荷在 q_1 处的电势

$$\varphi'_1 = \frac{q'_1}{4\pi\epsilon_0 (d'_1 - d_1)} = \frac{-2q}{4\pi\epsilon_0 (2R - \frac{R}{2})} = -\frac{q}{3\pi\epsilon_0 R}$$

$$\Rightarrow W_{1a} = \frac{q^2}{6\pi\epsilon_0 R}$$

球壳内表面的感应电荷与 q_1 之间的相互作用能.

$$W_{1b} = q_1 \varphi'_1 = -\frac{q^2}{3\pi\epsilon_0 R}$$

注: 空腔内静电能 $W_1 = W_{1a} + W_{1b} = -\frac{q^2}{6\pi\epsilon_0 R}$

与 $W_1 = \frac{1}{2} q_1 \varphi'_1 = -\frac{q^2}{6\pi\epsilon_0 R}$ 结果一致.

(4) 球壳外表面感应电荷的自能

$$W_{2a} = \frac{1}{2} \oint \tilde{\varphi}'_2 dq'_2$$

其中 φ'_2 为外表面感应电荷在自身处电势.

$$V = \tilde{\varphi}'_2 + \varphi_2$$

其中 φ_2 为 q_2 在内表面的电势

$$\Rightarrow W_{2a} = \frac{1}{2} q_2 V - \frac{1}{2} \oint \varphi_2 dq'_2 = \frac{1}{2} q_2 V - \frac{1}{2} q_2 \varphi'_2$$

其中 φ'_2 为外表面感应电荷在 q_2 处的电势

$$\begin{aligned} \varphi'_2 &= \frac{q'_2}{4\pi\epsilon_0 (d_2 - d'_2)} + \frac{q''_2}{4\pi\epsilon_0 d_2} \\ &= \frac{-\frac{\sqrt{5}-1}{2}q}{4\pi\epsilon_0 ((\sqrt{5}+1)R - (\sqrt{5}-1)R)} + \frac{\frac{\sqrt{5}+1}{2}q}{4\pi\epsilon_0 (\sqrt{5}+1)R} \\ &= \frac{3-\sqrt{5}}{4} \frac{q}{4\pi\epsilon_0 R} \end{aligned}$$

$$W_{2a} = \frac{1}{2} q \cdot \frac{\sqrt{5}+1}{4} \frac{q}{4\pi\epsilon_0 R} - \frac{3-\sqrt{5}}{8} \frac{q^2}{4\pi\epsilon_0 R} = \frac{\sqrt{5}-1}{4} \frac{q^2}{4\pi\epsilon_0 R}$$

球壳外表面的感应电荷与 q_2 之间的相互作用能.

$$W_{2b} = q_2 \varphi'_2 = \frac{3-\sqrt{5}}{4} \frac{q^2}{4\pi\epsilon_0 R}$$

注: 球壳外静电能 $W_2 = W_{2a} + W_{2b} = \frac{q^2}{8\pi\epsilon_0 R}$

$$\text{与 } W_2 = \frac{1}{2} q_2 V + \frac{1}{2} q_2 \varphi'_2 = \frac{q^2}{8\pi\epsilon_0 R}$$

(5) 空腔内静电能不变. 初、末态空腔外静电能为

$$W_2 = \frac{q^2}{8\pi\epsilon_0 R}$$

$$W'_2 = \frac{1}{2} \frac{q^2}{4\pi\epsilon_0 (2R)} = \frac{q^2}{16\pi\epsilon_0 R}$$

$$\Rightarrow A_1 = W'_2 - W_2 = -\frac{q^2}{16\pi\epsilon_0 R}$$

(6) 空腔内静电能不变.

维持导体的电势 V 不变.

设 Q 为末态球壳外表面电荷量

$$V = \frac{Q}{4\pi\epsilon_0 (2R)} \Rightarrow Q = 8\pi\epsilon_0 R V = \frac{\sqrt{5}+1}{2} q.$$

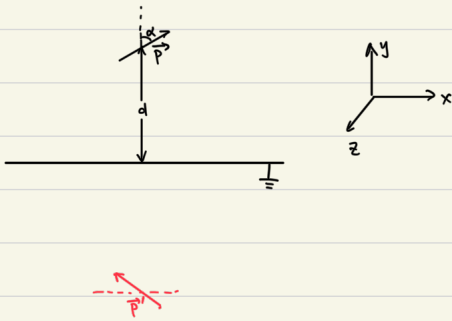
$$\text{末态空腔外静电能为 } W'_2 = \frac{1}{2} \frac{Q^2}{4\pi\epsilon_0 (2R)} = \frac{3+\sqrt{5}}{8} \frac{q^2}{4\pi\epsilon_0 R}$$

$$A'_2 + A_2 = W'_2 - W_2$$

$$\text{电源做功 } A'_2 = (Q - q)V = \frac{q^2}{8\pi\epsilon_0 R}$$

$$\Rightarrow A_2 = \frac{3+\sqrt{5}}{8} \frac{q^2}{4\pi\epsilon_0 R} - \frac{q^2}{8\pi\epsilon_0 R} - \frac{q^2}{8\pi\epsilon_0 R} = -\frac{5-\sqrt{5}}{8} \frac{q^2}{4\pi\epsilon_0 R}$$

例. 一个具有电偶极矩 $\vec{p}$ 的电偶极子位于无限大接地导体平面上方距离为 d 处. 该电偶极子与平面的法线的夹角为 α . 求该电偶极子受到的作用力和力矩.



$$\vec{p} = p(\sin\alpha \hat{x} + \cos\alpha \hat{y})$$

$$\vec{p}' = p(-\sin\alpha \hat{x} + \cos\alpha \hat{y})$$

$$\vec{R} = 2d \hat{y}$$

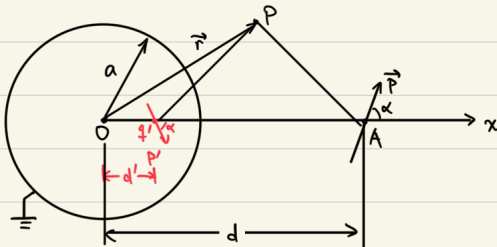
$$\begin{aligned} W &= 0 - \frac{1}{2} \vec{p} \cdot \vec{E}' = -\frac{1}{2} \vec{p} \cdot \left[\frac{1}{4\pi\epsilon_0} \left(\frac{3(\vec{p}' \cdot \vec{R})\vec{R}}{R^5} - \frac{\vec{p}'}{R^3} \right) \right] \\ &= -\frac{1}{8\pi\epsilon_0} \left[\frac{3(\vec{p}' \cdot \vec{R})(\vec{p} \cdot \vec{R})}{R^5} - \frac{\vec{p}' \cdot \vec{p}}{R^3} \right] \\ &= -\frac{1}{8\pi\epsilon_0} \left[\frac{3p^2 \cos^2\alpha}{(2d)^3} - \frac{p^2(\cos^2\alpha - \sin^2\alpha)}{(2d)^3} \right] \\ &= -\frac{p^2(1 + \cos^2\alpha)}{64\pi\epsilon_0 d^3} \end{aligned}$$

$$\vec{F} = -\nabla W|_{y=d} = -\frac{3p^2(1 + \cos^2\alpha)}{64\pi\epsilon_0 d^4} \hat{y}$$

$$\tau = -\frac{\partial W}{\partial \alpha}|_{y=d} = +\frac{p^2 \sin\alpha \cos\alpha}{32\pi\epsilon_0 d^4}$$

$$\vec{\tau} = \frac{p^2 \sin\alpha \cos\alpha}{32\pi\epsilon_0 d^4} (-\hat{z})$$

例. 半径为 a 的接地导体球外 A 点处有一电偶极子 $\vec{p}$, A 到球心 O 的距离为 d , $\vec{p}$ 与 OA 夹角为 α .



(1) 求导体球外的静电势.

(2) 将电偶极子缓慢移至无穷远处, 外界需要做多少功?

(3) 求电偶极子受力.

$$\begin{aligned} \vec{p}: +q \quad r_+ &= \sqrt{d^2 + \left(\frac{a}{2}\right)^2 + 2d \cdot \frac{a}{2} \cos\alpha} \\ -q \quad r_- &= \sqrt{d^2 + \left(\frac{a}{2}\right)^2 - 2d \cdot \frac{a}{2} \cos\alpha} \end{aligned}$$

$$(+q)' = -\frac{q}{r_+} \approx -\frac{q}{d} \left(1 - \left(\frac{a}{d}\right) \cos\alpha\right)$$

$$(-q)' = +\frac{q}{r_-} \approx +\frac{q}{d} \left(1 + \left(\frac{a}{d}\right) \cos\alpha\right)$$

$$d_+ = \frac{a}{r_+} \approx \frac{a}{d^2} \left(1 - 2 \cdot \left(\frac{a}{d}\right) \cos\alpha\right)$$

$$d_- = \frac{a}{r_-} \approx \frac{a}{d^2} \left(1 + 2 \cdot \left(\frac{a}{d}\right) \cos\alpha\right)$$

$$\vec{p}' = +\frac{q}{d} \cdot \left(\frac{a}{d}\right)^2 l \left(\cos\alpha \hat{x} - \sin\alpha \hat{y}\right)$$

$$q' = +\frac{q}{d} \cdot \frac{a}{d} \cos\alpha$$

$$\vec{p} \text{ 的镜像为 } d' = \frac{a^2}{d} \text{ 处. } \vec{p}' = \frac{a^3}{d^3} p \left(\cos\alpha \hat{x} - \sin\alpha \hat{y}\right)$$

$$q' = \frac{a^2 p}{d^2} \cos\alpha$$

$$(1) \quad \varphi(\vec{r}) = \frac{\vec{p} \cdot (\vec{r} - \vec{d})}{4\pi\epsilon_0 |\vec{r} - \vec{d}|^3} + \frac{\vec{p}' \cdot (\vec{r} - \vec{d}')}{4\pi\epsilon_0 |\vec{r} - \vec{d}'|^3} + \frac{q'}{4\pi\epsilon_0 |\vec{r} - \vec{d}'|}$$

$$(2) \quad \vec{E}' = \frac{1}{4\pi\epsilon_0} \left[\frac{3(\vec{p}' \cdot \vec{R})\vec{R}}{R^5} - \frac{\vec{p}'}{R^3} + q' \frac{\vec{R}}{R^3} \right] \text{ 其中 } \vec{R} = \vec{d} - \vec{d}' = (d-d') \hat{x}$$

$$\begin{aligned} W &= -\frac{1}{2} \vec{p} \cdot \vec{E}' = -\frac{1}{8\pi\epsilon_0} \left[\frac{3(\vec{p}' \cdot \vec{R})(\vec{p} \cdot \vec{R})}{R^5} - \frac{\vec{p}' \cdot \vec{p}}{R^3} + q' \frac{\vec{p} \cdot \vec{R}}{R^3} \right] \\ &= -\frac{1}{8\pi\epsilon_0} \left[\frac{3 \frac{a^3}{d^3} p \cos\alpha \cdot p \cos\alpha}{(d - \frac{a^2}{d})^3} - \frac{\frac{a^3}{d^3} p^2 (\cos^2\alpha - \sin^2\alpha)}{(d - \frac{a^2}{d})^3} + q' \frac{p \cos\alpha}{(d - \frac{a^2}{d})^2} \right] \\ &= -\frac{1}{8\pi\epsilon_0} \frac{3a^3 p^2 \cos^2\alpha - a^3 p^2 \cos^2\alpha + a^3 p^2 \sin^2\alpha + a p^2 \cos^2\alpha \cdot (d^2 - a^2)}{(d^2 - a^2)^3} \\ &= -\frac{p^2 a (a^2 + d^2 \cos^2\alpha)}{8\pi\epsilon_0 (d^2 - a^2)^3} \end{aligned}$$

$$\Rightarrow A = 0 - W = \frac{p^2 a (a^2 + d^2 \cos^2\alpha)}{8\pi\epsilon_0 (d^2 - a^2)^3}$$

$$(3) \quad \vec{F} = -\nabla W$$

$$F_x = -\frac{\partial W}{\partial x}|_{x=d} = -\frac{a d p^2 [3a^2 + (2d^2 + a^2) \cos^2\alpha]}{4\pi\epsilon_0 (d^2 - a^2)^4}$$

$$\text{沿 } y \text{ 轴移动 } \Delta y \text{ 等效于相对球心旋转 } \Delta\theta \approx \frac{\Delta y}{d}. \quad \alpha' = \alpha - \Delta\theta = \alpha - \frac{\Delta y}{d}$$

$$F_y = -\frac{\partial W}{\partial y}|_{x=d} = -\frac{\partial W}{\partial \alpha} \cdot \frac{\partial \alpha}{\partial y}|_{x=d} = -\frac{\partial W}{\partial \alpha}|_{x=d} \left(-\frac{1}{d}\right) = \frac{a d p^2 \sin\alpha \cos\alpha}{4\pi\epsilon_0 (d^2 - a^2)^3}$$

$$\Rightarrow \vec{F} = \frac{a d p^2}{4\pi\epsilon_0 (d^2 - a^2)^4} \left[-(3a^2 + (2d^2 + a^2) \cos^2\alpha) \hat{x} + (d^2 - a^2) \sin\alpha \cos\alpha \hat{y} \right]$$

$$\text{检验. } d_0 = d - a, \quad a \rightarrow \infty$$

$$W = -\frac{p^2(1 + \cos^2\alpha)}{64\pi\epsilon_0 d^3}$$

$$\vec{F} = -\frac{3p^2(1 + \cos^2\alpha)}{64\pi\epsilon_0 d^4} \hat{y}$$

与前一例题中结果相同.